

TriZOD — Final Pipeline & Re-Referenced Dataset

Step 8 wildcards · LACS in scoring · .str emission · alpha-synuclein

Tobias Senoner

6 May 2026

TUM · TriZOD project meeting

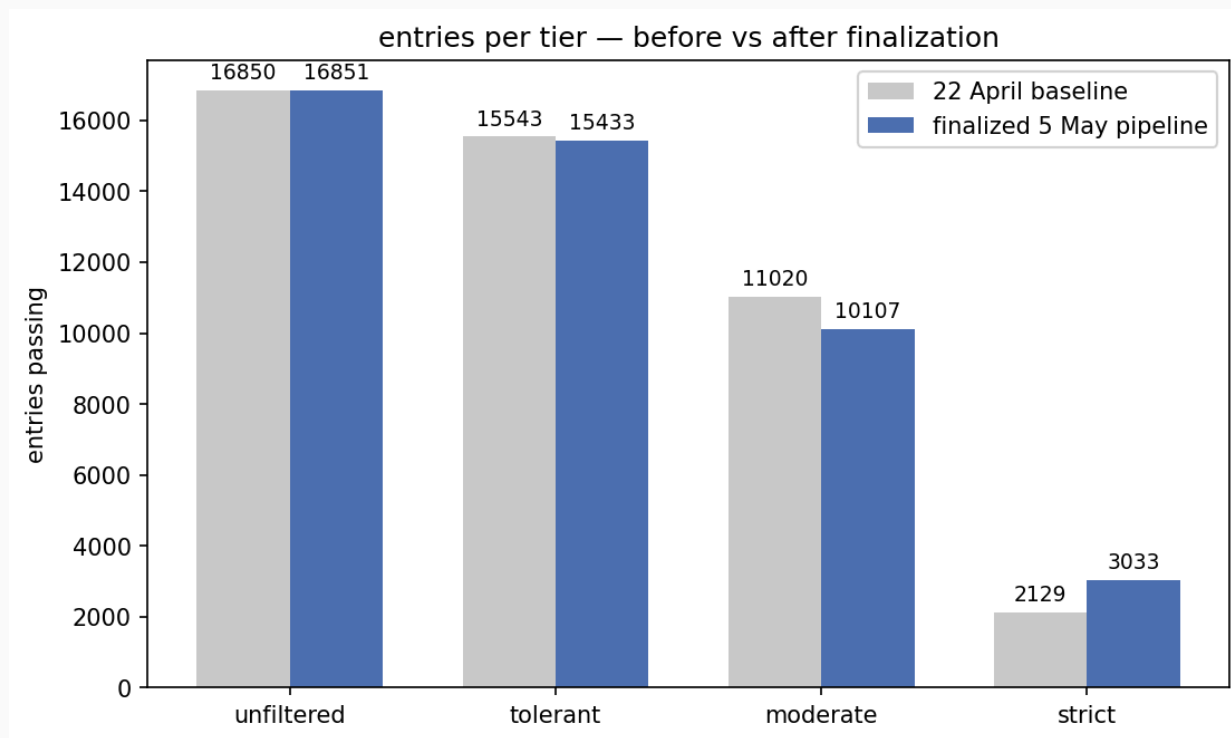
0.1 TriZOD in one slide

- **Goal:** per-residue disorder scores from BMRB NMR shifts (CheZOD-style)
- **Inputs:** 17,388 BMRB NMR-STAR entries
- **Output:** per-tier filtered JSON/CSV + (new) re-referenced `.str` files
- **Z-score:** deviation from POTENCI random-coil prediction
- **G-score:** $[0, 1]$ geometric-mean form combining backbone atom Z-scores

0.2 What's new since 22 April

- **Step 8:** Leu/Val ambiguous methyl wildcards (CDx / CGx)
- **Step 9:** LACS pre-correction baked into the scoring pipeline
- New flag: `--rereference-mode {none, lacs, potenci-only, both}` (default both)
- New flag: `--emit-str <dir>` emits per-entry re-referenced NMR-STAR files
- **Zenodo deposit metadata** in repo (DOI on first tagged release)

0.3 Filter improvements (Steps 4–7) and per-tier dataset deltas

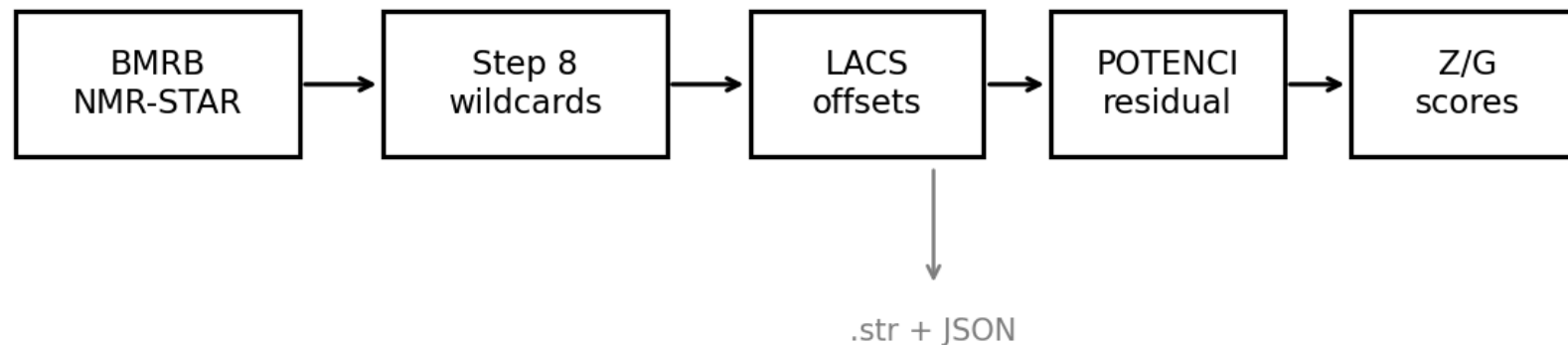


Removed denaturant false-positives, fixed solid-state regex, added paramagnetic exclusion, relaxed min-backbone-shift-types 5→4 (strict), broadened the Celsius heuristic.

0.4 Step 8: methyl wildcards

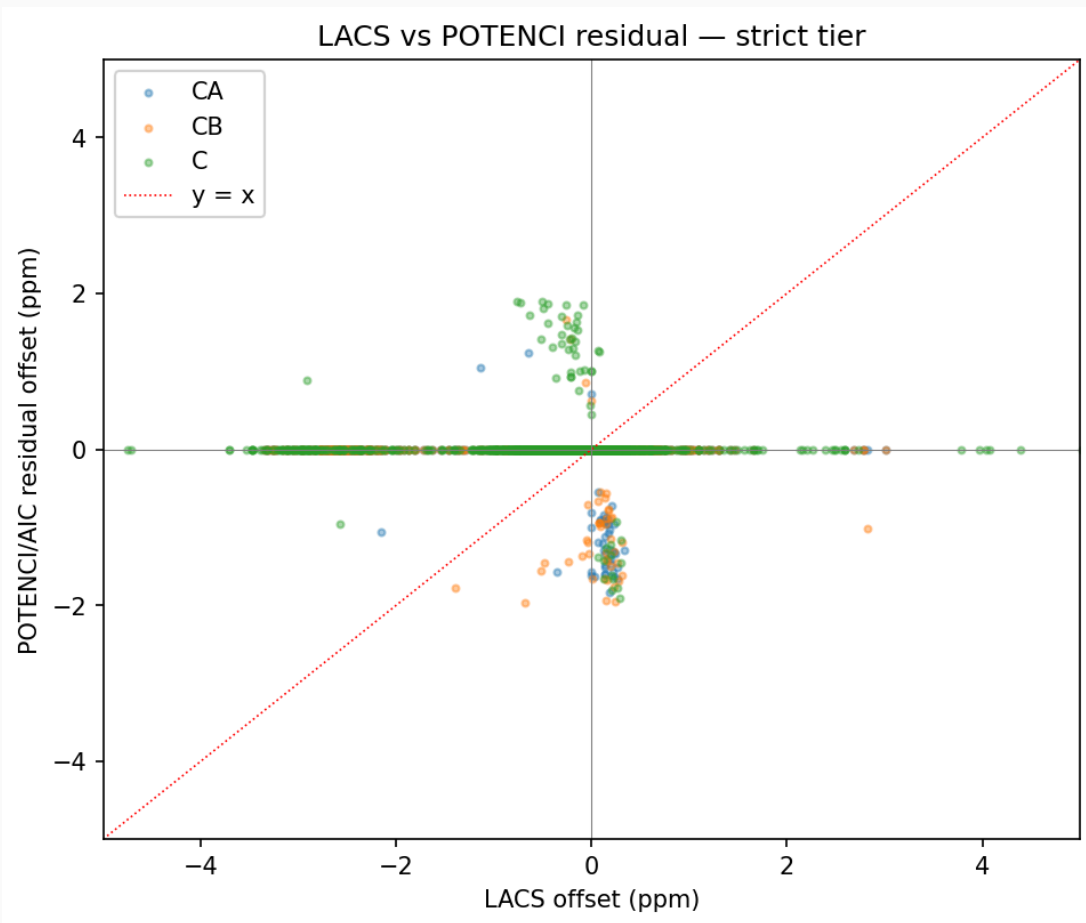
- LEU CD1/CD2 -> CDx, VAL CG1/CG2 -> CGx for non-stereospecific ambiguity codes
- Stereospecific assignments (ambiguity code 1) preserved as-is
- **Backbone scoring unchanged** — wildcards surface only in the emitted .str files
- Restores otherwise-discarded sidechain methyl shifts for downstream consumers

0.5 Re-referencing in the pipeline



raw shifts → **LACS pre-correction** (Wishart RC tables, robust line fits) → **POTENCI residual** (AIC-gated rolling 9-window) → Z / G-scores

0.6 LACS vs POTENCI residual capture

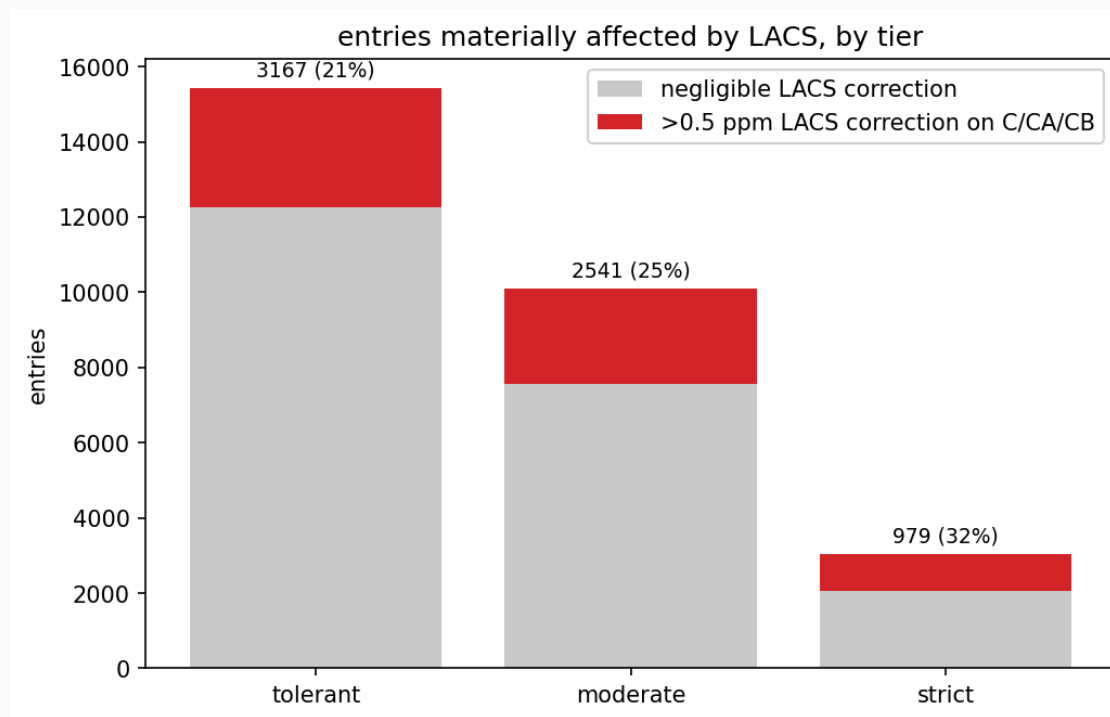


Per-entry CA / CB / C offsets in the strict tier.

LACS catches the large systematic referencing bias.

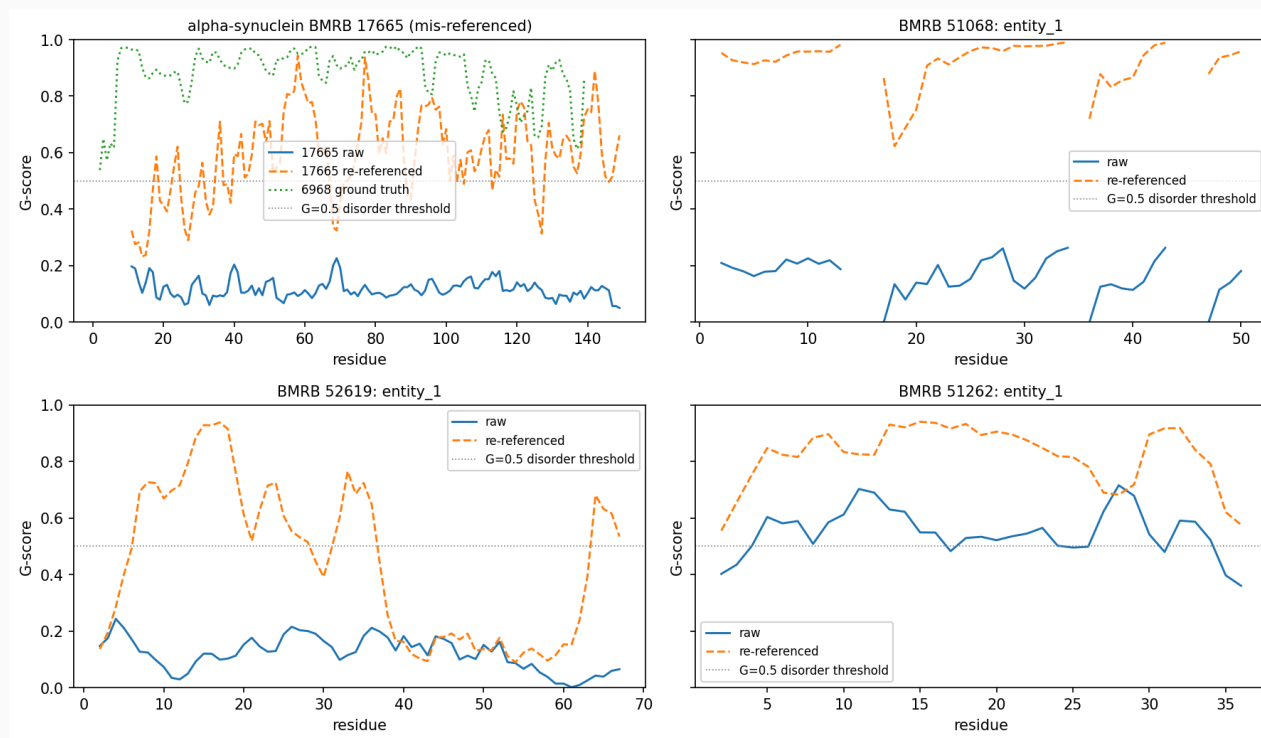
POTENCI/AIC residual captures the remainder — most entries land near the origin once LACS has done its job.

0.7 Entries materially affected by LACS, by tier



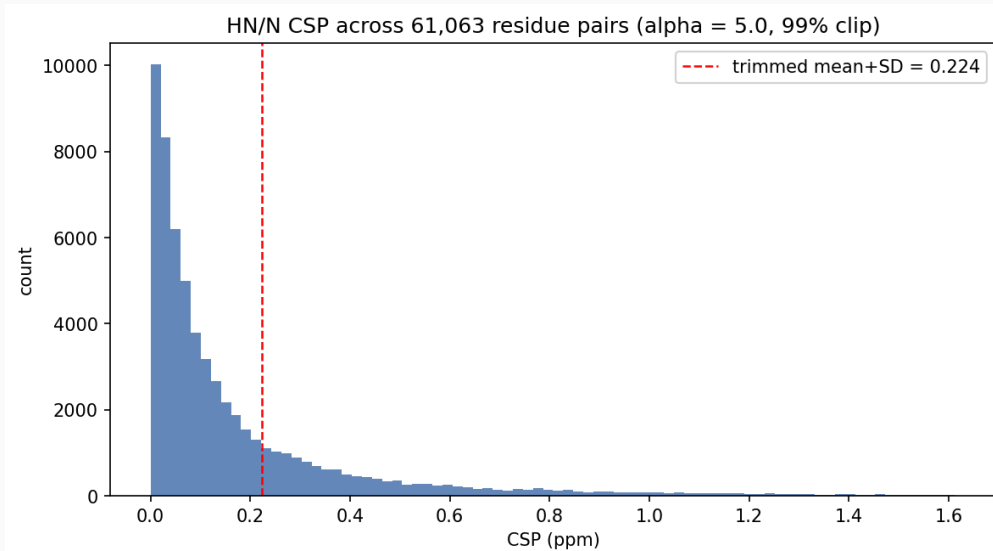
Red = entries with $|\text{LACS offset}| > 0.5$ ppm on at least one of C/CA/CB. Re-referencing meaningfully changes the input shifts for 21% (tolerant), 25% (moderate), 32% (strict).

0.8 Reid #2: alpha-synuclein and top-3 G-score flippers

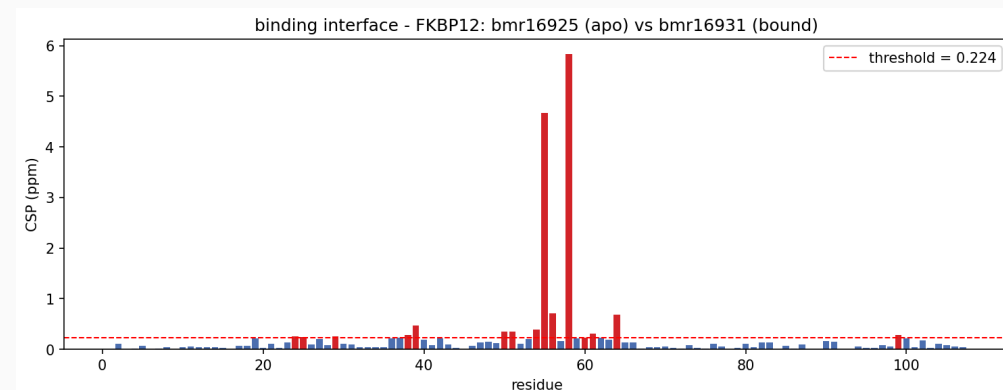


BMRB 17665 raw → looks helical (matches original helical-tetramer paper); re-referenced → disordered, matching BMRB 6968 ground truth. α Syn LACS offset CA/CB = +2.82 ppm. Top-3 flippers 51068/52619/51262 show the same pattern.

0.9 Reid #1: chemical shift perturbations on bound/unbound pairs

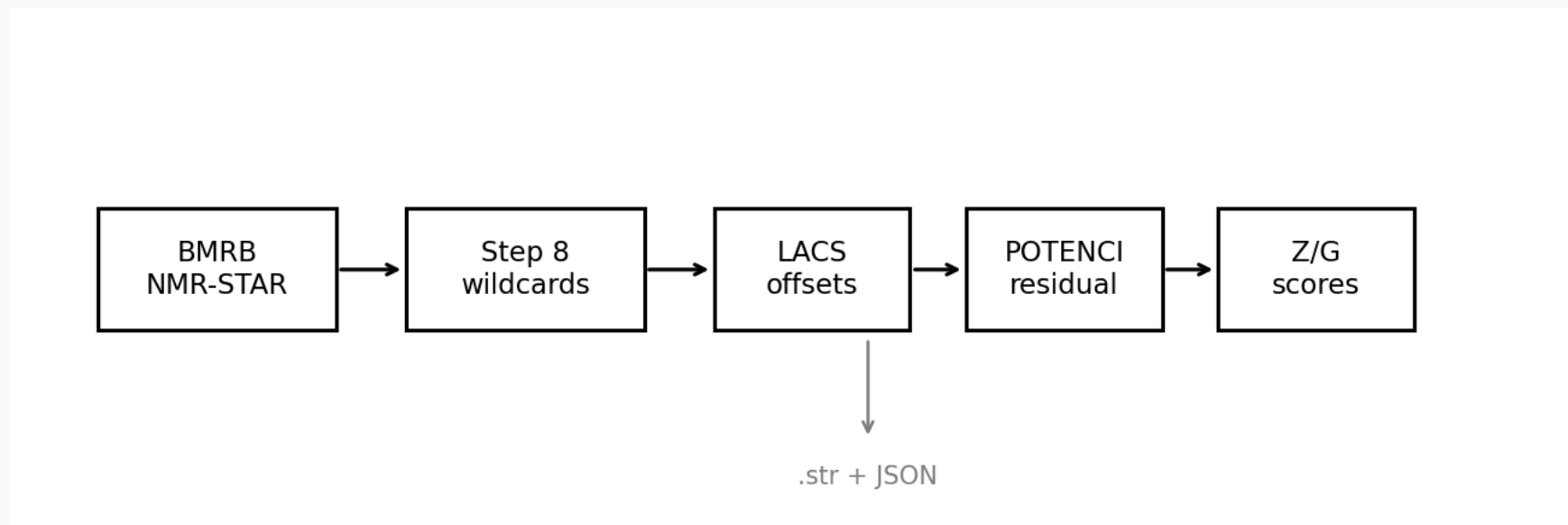


CSP across 581 pairs · 61,063 values · threshold 0.224 ppm



FKBP12 apo (bmr16925) vs bound (bmr16931) — interface at res. 55, 58

0.10 Final pipeline architecture



- `--rereference-mode {none, lacs, potenci-only, both}`
- `--emit-str` writes one `.str` per scored entity (two saveframes)
- JSON now carries both `lacs_off_<atom>` and `off_<atom>`
- All four filter tiers re-run on the finalized pipeline

0.11 Released artefacts

- `data/release/<tier>/scores.json` — per-residue Z/G + LACS + POTENCI offsets
- `data/release/<tier>/str/bmr*_*_rereferenced.str` — re-referenced NMR-STAR
- `.zenodo.json` + `CITATION.cff` — DOI on first tagged release

Per-tier counts after rerun:

Tier	Entries	Coverage
strict	3,033	17.00%
moderate	10,107	56.64%
tolerant	15,433	86.49%
unfiltered	16,851	94.45%

0.12 Next steps

- Push deposit to Zenodo (DOI placeholder until first tag)
- Iterate on Reid's CSP — extend to multi-atom $\Delta\omega_{\text{RMS}}$ (Schumann/Williamson 2008)
- Step 8 downstream: any auto-assignment partner who'd benefit from CDx/CGx?
- Reid follow-up with John Markley on why BMRB stopped LACS reports (frozen July 2020)